

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

14

10 questions · ~15 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS

In a box of pens, the ratio of black pens to red pens is 8 to 1. There are 40 black pens in the box. How many red pens are in the box?

- A. 5
- B. 8
- C. 40
- D. 320

Q2

GRID-IN

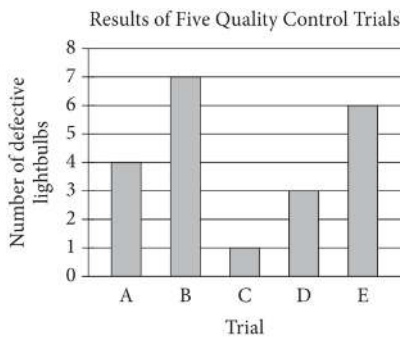
PROBLEM-SOLVING AND DATA ANALYSIS

There are 170 blocks in a container. Of these blocks, 10% are green. How many blocks in the container are green?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

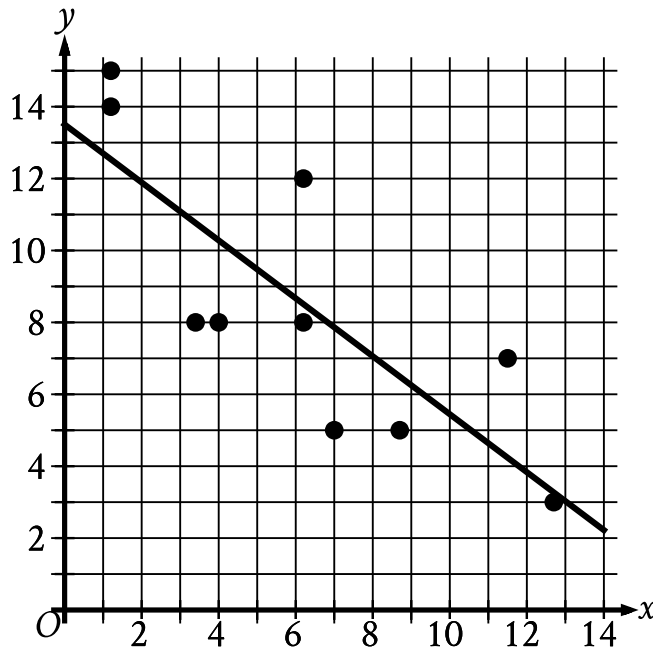
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



For quality control, a company that manufactures lightbulbs conducted five different trials. In each trial, 500 different lightbulbs were tested. The bar graph above shows the number of defective lightbulbs found in each trial. What is the mean number of defective lightbulbs for the five trials?

- A. 4.0
- B. 4.2
- C. 4.6
- D. 5.0

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
 - The line of best slants down from left to right.
 - 1 point is touching the line of best fit.
 - 4 points are above the line of best fit.
 - 5 points are below the line of best fit.
 - The line of best fit passes through the following approximate coordinates:
 - (2 comma 12)
 - (8 comma 7)
 - (13 comma 3)

Which of the following equations best represents the line of best fit shown?

- A. $y = 13.5 + 0.8x$
- B. $y = 13.5 - 0.8x$
- C. $y = -13.5 + 0.8x$
- D. $y = -13.5 - 0.8x$

Q5

PROBLEM-SOLVING AND DATA ANALYSIS

The ratio of the length of line segment XY to the length of line segment ZV is 6 to 1. If the length of line segment XY is 102 inches, what is the length, in inches, of line segment ZV ?

- A. 17
- B. 96
- C. 102
- D. 612

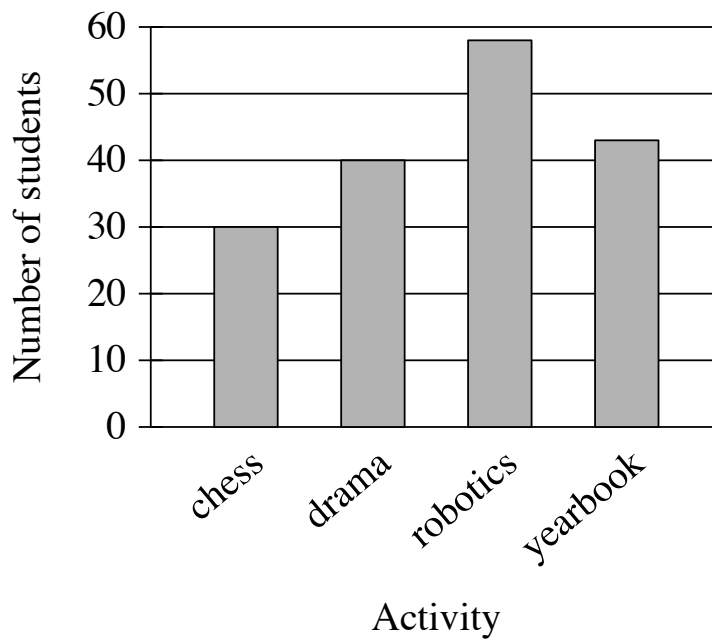
Q6

PROBLEM-SOLVING AND DATA ANALYSIS

What is 10% of 370?

- A. 27
- B. 37
- C. 333
- D. 360

The bar graph shows the distribution of the number of students in each of four extracurricular activities at a high school.



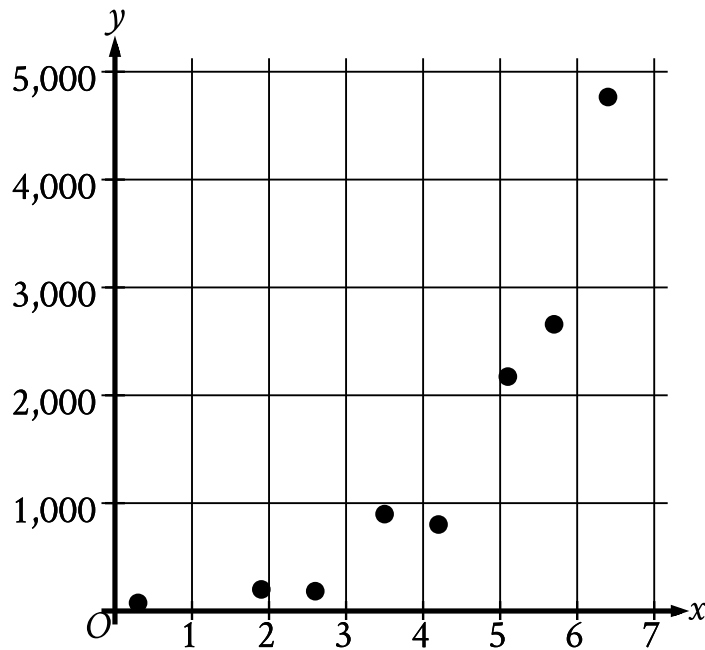
- The Number of students data for the 4 bars are as follows:
 - chess: 30
 - drama: 40
 - robotics: 58
 - yearbook: 43

How many more students are in drama than in chess?

- A. 10
- B. 30
- C. 40

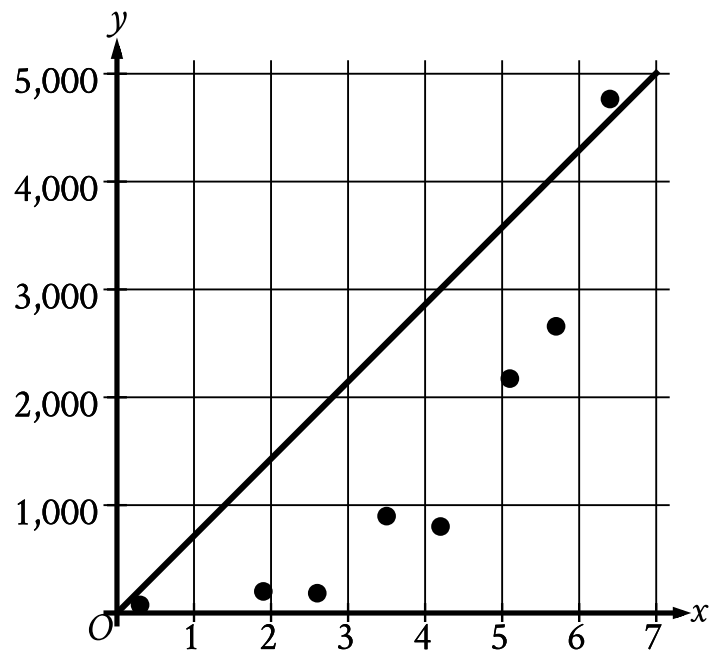
D. 70

The scatterplot shows the relationship between two variables, x and y .



- The scatterplot has 8 data points.
- The data points are in an exponential pattern trending up from left to right.
- The data points have the following approximate coordinates:
 - (0.3 comma 75)
 - (1.9 comma 200)
 - (2.6 comma 183)
 - (3.5 comma 898)
 - (4.2 comma 802)
 - (5.1 comma 2,173)
 - (5.7 comma 2,658)
 - (6.4 comma 4,766)

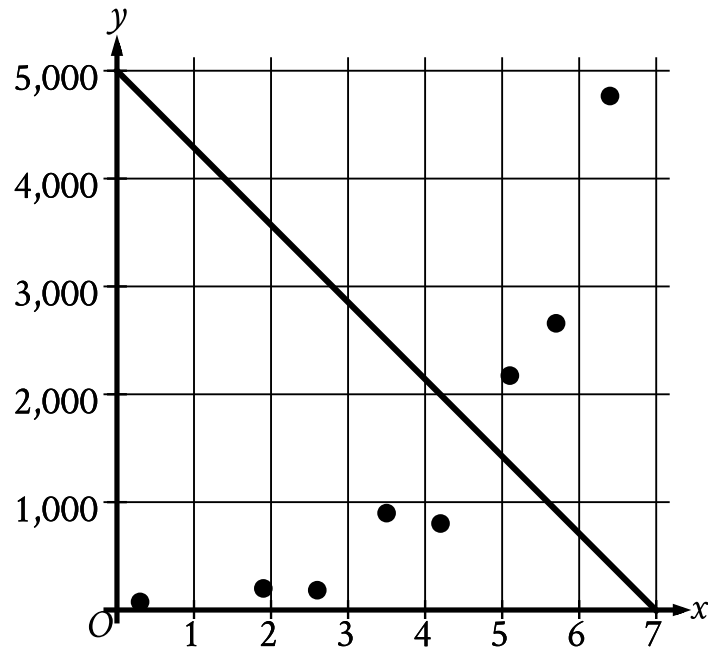
Which of the following graphs shows the most appropriate model for the data?



A.

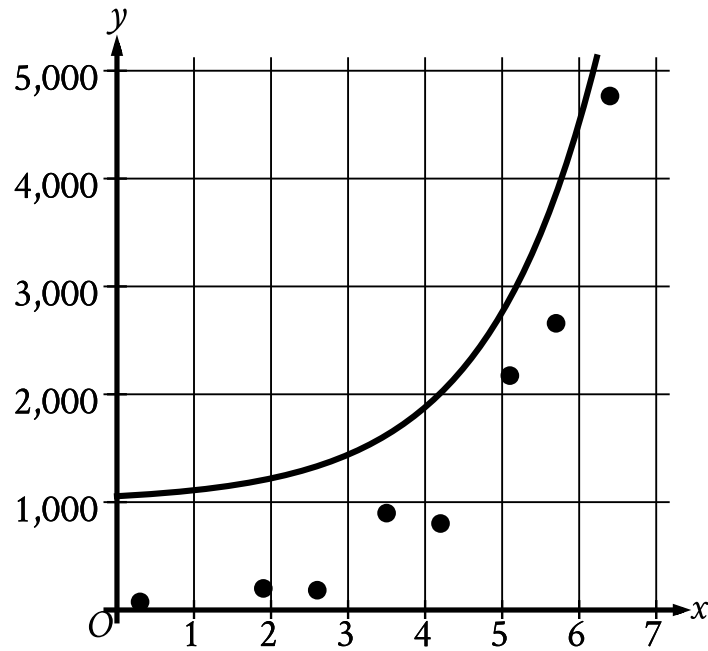
- The scatterplot has 8 data points.
- The data points are in an exponential pattern trending up from left to right.
- A line of best fit is shown:
 - The line of best fit slants up from left to right.
 - 0 points are touching the line of best fit.
 - 1 point is above the line of best fit.
 - 7 points are below the line of best fit.
 - The line of best fit passes through the following approximate coordinates:
 - (0 comma 0)
 - (6 comma 4,290)

B.



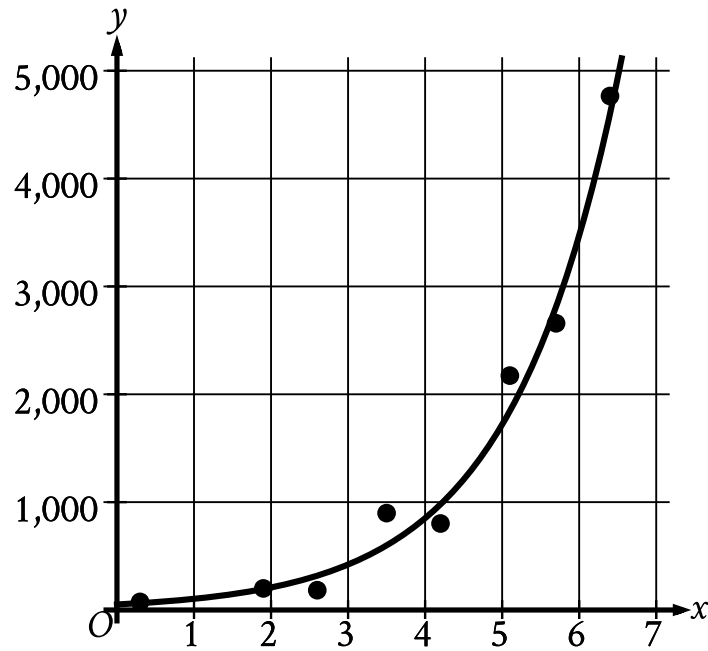
- The scatterplot has 8 data points.
- The data points are in an exponential pattern trending up from left to right.
- A line of best fit is shown:
 - The line of best fit slants down from left to right.
 - 0 points are touching the line of best fit.
 - 3 points are above the line of best fit.
 - 5 points are below the line of best fit.
 - The line of best fit passes through the following coordinates:
 - (0 comma 5,000)
 - (6 comma 710)

c.



- The scatterplot has 8 data points.
- The data points are in an exponential pattern trending up from left to right.
- A curve of best fit is shown:
 - The curve of best fit trends up from left to right.
 - 0 points are touching the curve of best fit.
 - 0 points are above the curve of best fit.
 - 8 points are below the curve of best fit.
 - The curve of best fit passes through the following coordinates:
 - (0 comma 1,055)
 - (4 comma 1,880)
 - (6 comma 4,520)

D.



- The scatterplot has 8 data points.
- The data points are in an exponential pattern trending up from left to right.
- A curve of best fit is shown:
 - The curve of best fit trends up from left to right.
 - 2 points are touching the curve of best fit.
 - 3 points are above the curve of best fit.
 - 3 points are below the curve of best fit.
 - The curve of best fit passes through the following approximate coordinates:
 - (0 comma 51)
 - (4 comma 851)
 - (6 comma 3,472)

Q9

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A product costs 11.00 dollars per pound. What is the cost, in dollars, for 6 pounds of the product?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10

PROBLEM-SOLVING AND DATA ANALYSIS

21 is 21% of what number?

- A. 0
- B. 1
- C. 42
- D. 100